Orbits, cycles and the alternating group

Fix σ a permutation of a set $A \Rightarrow$ it induces a partition on A.

- Given $a, b \in A$ they are in the same cell iff $b = \sigma^n(a)$ for some $n \in \mathbb{N}$.

For $a, b \in A$

$$a \sim_\sigma b \text{ iff } \exists n \in \mathbb{Z} \text{ s.t. } \sigma^n(a) = b.$$

$\sim_\sigma$ is an equivalence relation:

- reflexivity: $\sigma^0(a) = a$.
- symmetry: $\sigma^n(a) = b \Rightarrow a = \sigma^{-n}(b)$
- transitivity: $\sigma^n(a) = b$ and $\sigma^m(b) = c$

$$\Rightarrow \sigma^m(b) = \underbrace{\sigma^m(\sigma^n(a))}_{\sigma^{m+n}(a)} = c.$$

$$\sigma^{m+n}(a) = c.$$

Definition: Let σ be a permutation of a set A the equivalence classes of $\sim_\sigma$ are the ORBITS of σ.

Example: $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 7 & 4 & 1 & 5 & 2 \end{pmatrix}$

Find the orbits of σ:

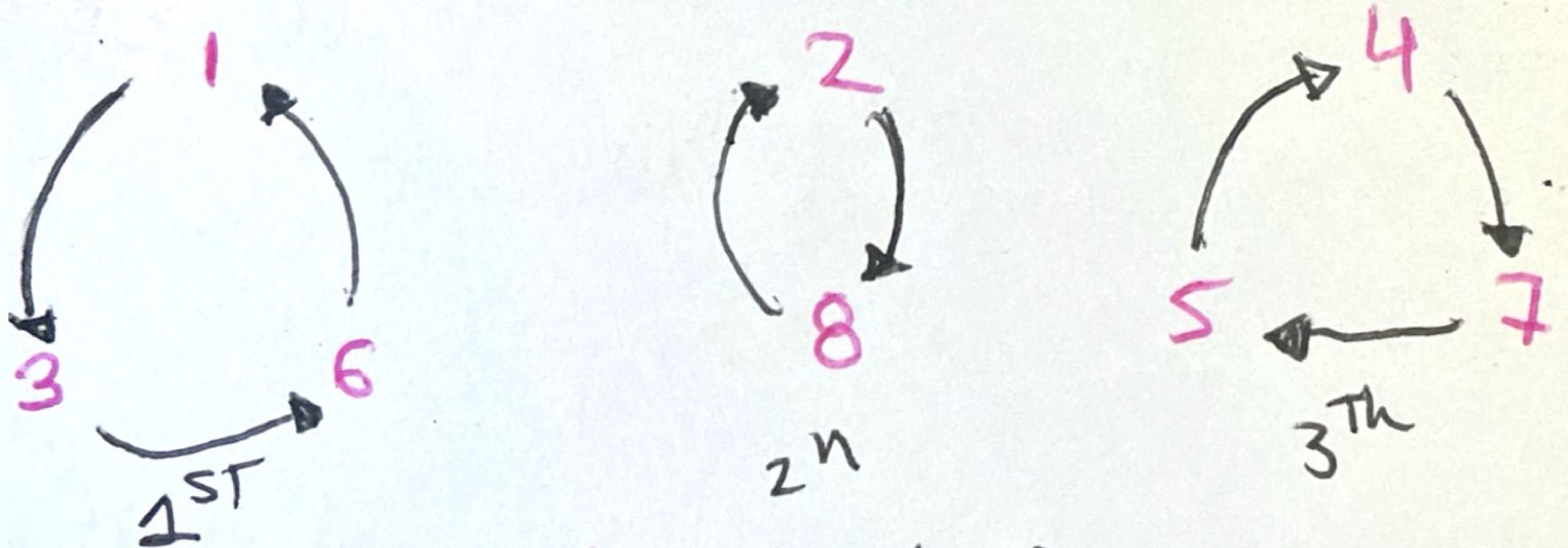
$1 \to 3 \to 6 \to 1$ $\qquad\{1,3,6\}$

$2 \rightleftarrows 8$ $\qquad\{2,8\}$

$4 \to 7 \to 5 \to 4$ $\qquad\{4,5,7\}$.

CYCLES:

We can represent the above graphically.



Each of them could be a single permutation.

$\begin{pmatrix} \underline{1} & 2 & \underline{3} & 4 & 5 & \underline{6} & 7 & 8 \\ \underline{3} & 2 & \underline{6} & 4 & 5 & \underline{1} & 7 & 8 \end{pmatrix}$ and the rest is the same.

↑ is a cycle → one orbit of size 3 $\{1,3,6\}$

5 of size 1 $\{2\}\{4\}\{5\}\{7\}\{8\}$

Definition: A permutation is a cycle if it has at most one orbit containing more than one element. The length of a cycle is the number of elements in its largest orbit.

↳ **new notation:** we write $(1,3,6)$

Example: Working in S_5 we see that:

$$(1,3,5,4) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 1 & 4 \end{pmatrix}$$

Our main theorem: Every permutation σ of a finite set is a product of disjoint cycles.

In our example:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 7 & 4 & 1 & 5 & 2 \end{pmatrix} = (1,3,6)(2,8)(4,7,5)$$

disjoint: any element is moved by at most one of these cycles.

Proof: Let $B_1, B_2, \ldots, B_r$ be the orbits of σ and let

$$\mu_i(x) = \begin{cases} \sigma(x) & \text{if } x \in B_i, \\ x & \text{otherwise.} \end{cases}$$

then $\sigma = \mu_1 \cdots \mu_r$. Since the classes are disjoint so are the cycles.

Why is this useful?

- multiplication of permutations is not commutative
- multiplication of disjoint cycles is commutative.

Remark: Since the orbits of a permutation are unique, the representation of a permutation as a product of disjoint cycles is unique up to order of factors.

Example:

Consider $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 2 & 4 & 3 & 1 \end{pmatrix}$

$1 \to 6 \to 1$ $\quad (1,6)$

$2 \to 5 \to 3$ $\quad (2,5,3)$

$(1,6)(2,5,3) = (2,5,3)(1,6)$

Even and odd Permutations:

Definition: A cycle of length 2 is a <u>transposition</u>.

leaves all elements but 2 fixed and maps one onto the other.

<u>Remark:</u> Any cycle can be decompose as a product of transpositions.

e.g. $(2,5,3) = \underbrace{(2,3)}_{2^{nd}} \underbrace{(2,5)}_{1^{ST}}$

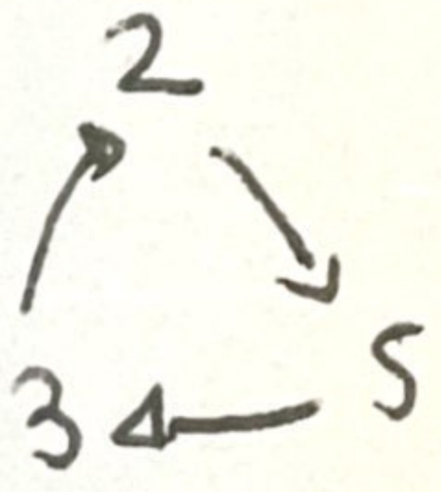
~~2 3 5~~

~~3 2 5 1^{ST}~~

2 3 5

5 3 2 $\quad 1^{ST}$

5 2 3 $\quad 2^{nd}$.



In general:

$$(a_1, \dots, a_n) = (a_1 a_n)(a_1 a_{n-1}) \dots (a_1, a_2)$$

$\boxed{\text{Corollary}}$ Any permutation of a finite set of at least 2 elements is a product of transpositions.

e.g the $\sigma = (1,6)(2\ 5\ 3)$

$= (1,6)(2,3)(2,5)$

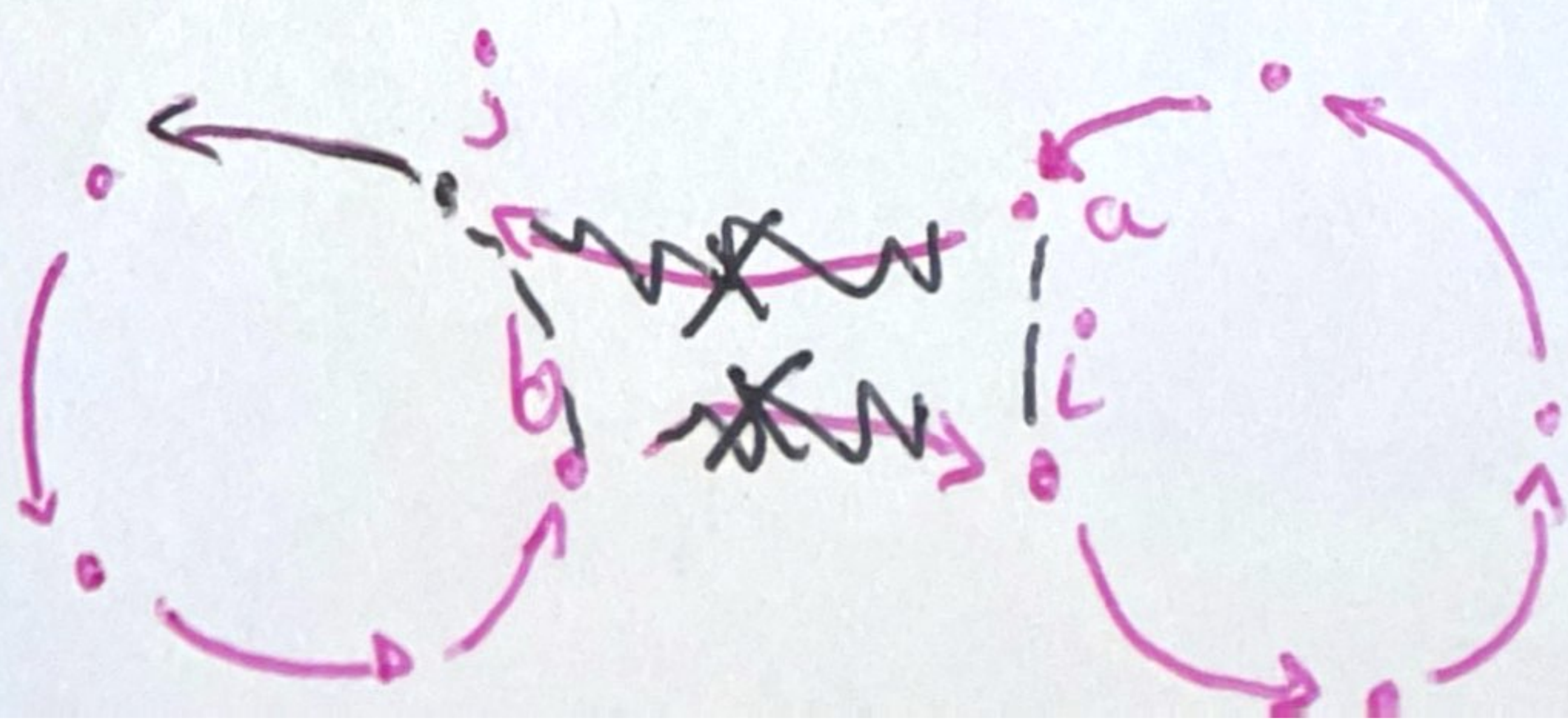
Warning: they are NOT longer disjoint.

Theorem: No permutation in S_n can be expressed both as a product of an even number of transpositions and as a product of an odd number of transpositions.

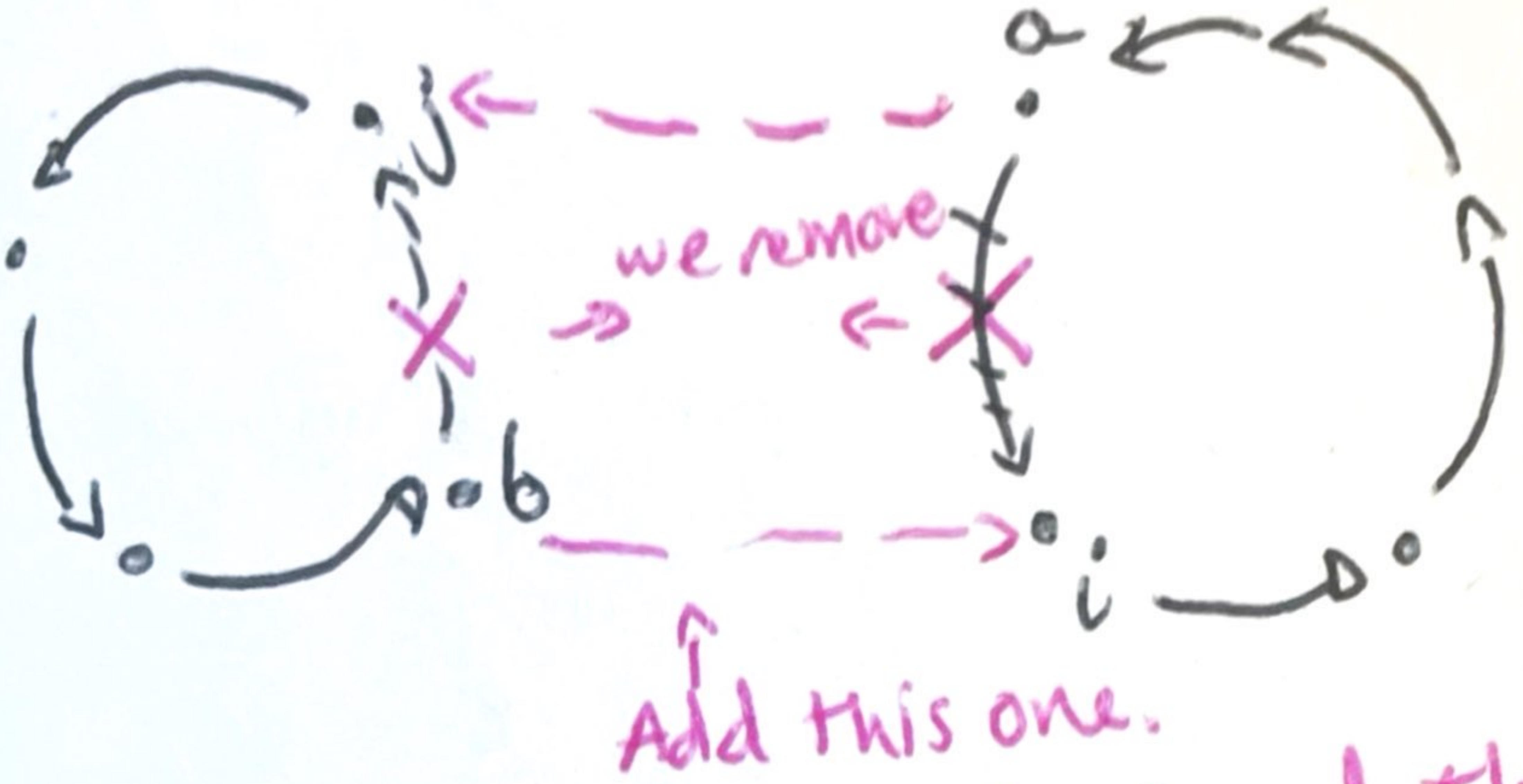
Proof: let $\sigma \in S_n$ and let $\tau = (i,j)$ be a transposition in S_n. We claim that the number of orbits of σ and $\tau\sigma$ differ by 1.

CASO 1: suppose i and j are in different orbits of σ

write σ as a product of disjoint cycles the first one that contains j and the second one that contains i.



More concretely we had 2 different cycles.



when we compute the product of the first 3 cycles in $\tau\sigma = (i,j)\sigma$

y tenemos

$(i,j)(b,j,x,x,x)(a,i,\otimes,\otimes,\otimes) =$

$(a,j,x,x,x,b,i,\otimes,\otimes,\otimes)$

Then the # of orbits of $\tau\sigma$ differs from the # of orbits of σ by 1.

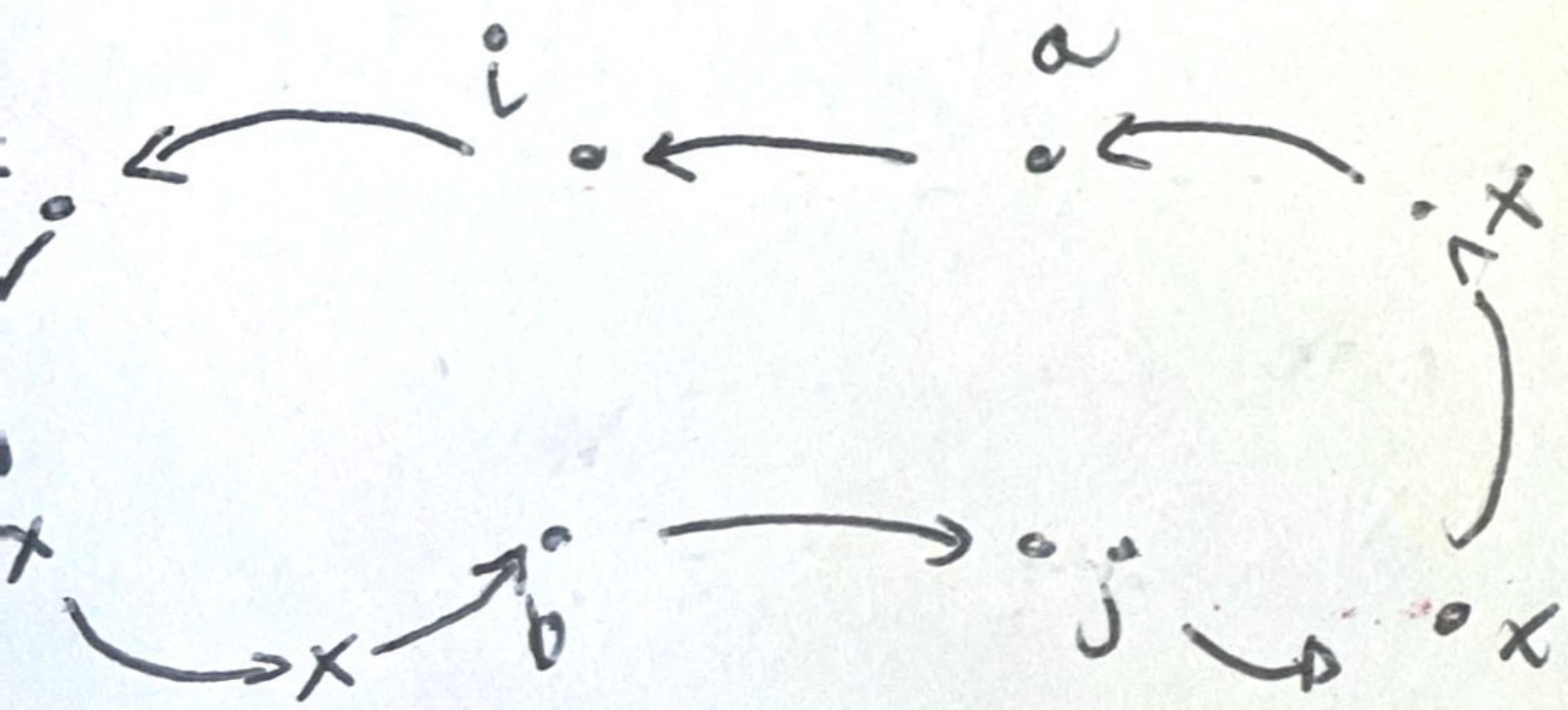
since the original 2 orbits have been glued into a single one.

$\Rightarrow$ Ex: the same happens if i and j are their only element in the orbit.

And

CASO 2: suppose i and j are in the same orbit of σ. We can write σ as a product of disjoint cycles with the first one of the form

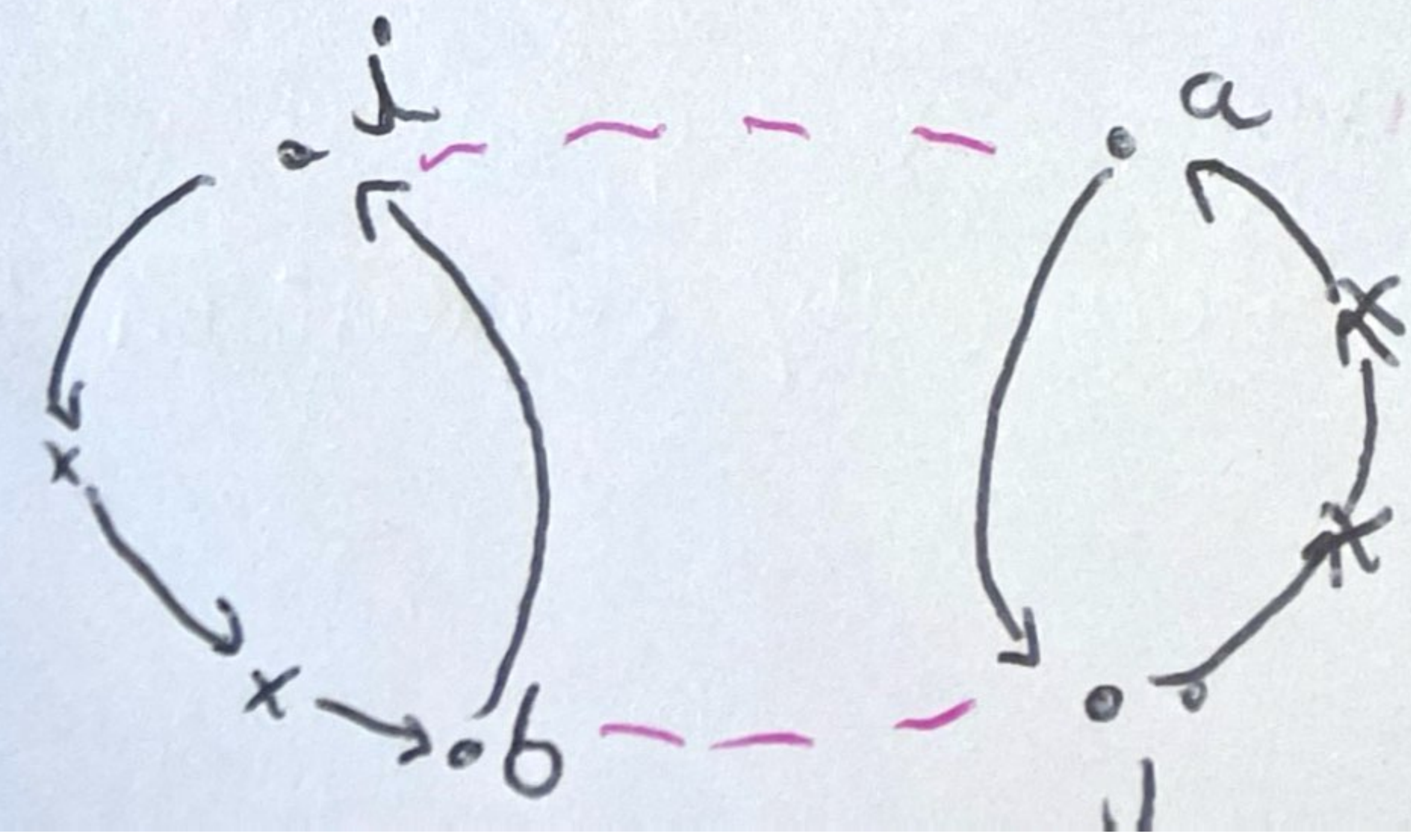
$(a, i, x, x, x, b, j, x, x)$ shown symbolically in the picture.



ow when we compute the product

the first 2 cycles $\tau\sigma$ in

$, j)\,\sigma$ we split in in 2 orbits.



- Thus $\tau\sigma$ has 1 more orbit than σ
- The identity permutation i has n-orbits because each element is the only element of its orbit.
- Number of orbits of a given permutation $\sigma \in S_n$ differs from n by either an even number or an odd number, but <u>not both</u>.

$$\sigma = \tau_1 \tau_2 \tau_3 \ldots \tau_m i$$

where τ_k are transpositions in 2 ways, once with m even and once with m odd.

<u>Definition</u>: A permutation of a finite set is even or odd according to whether it can be expressed as a product of an even number of transpositions or the product of an odd number of transpositions.

The Alternating group:

For $n \geq 2$ we want to study S_n

- # of even permutations = # of odd permutations $= \frac{n!}{2}$.

Let τ be a fixed transposition in S_n it exists since $n \geq 2$, we may as well suppose that $\tau = (1,2)$ we define the function:

$$\lambda_\tau : \underset{\text{even}}{A_n} \longrightarrow \underset{\text{odd}}{B_n}$$

$$\sigma \longrightarrow \tau\sigma$$

$$\sigma \longrightarrow (1,2)\sigma$$

σ is even $\Rightarrow$ $(1,2)\sigma$ is a product of 1 + even number of transpositions // or odd number which is therefore in B_n.

claim1: λ_τ is injective

if $\lambda_\tau(\sigma_1) = \lambda_\tau(\sigma_2)$ then $(1,2)\sigma_1 = (1,2)\sigma_2$

since S_n is a group

$$(1,2)^{-1}((1,2)\sigma_1) = (1,2)^{-1}((1,2)\sigma_2)$$

then $\boxed{\sigma_1 = \sigma_2}$

claim2: λ_τ is surjective.

And note that $\tau = \tau^{-1} = (1,2)$

thus if $\rho \in B_n$ then $\tau^{-1}\rho \in A_n$ and

$$\lambda_\tau(\tau^{-1}\rho) = \tau(\tau^{-1}\rho) = \rho.$$

thus λ_τ is a one to one correspondance.

Theorem: If $n \geq 2$ the collection of all even permutations of $\{1, 2, \ldots, n\}$ forms a subgroup of order $\frac{n!}{2}$ of the symmetric group.

identity: $e = (1,2) \circ (1,2)$

closed under $*$: $\underset{\substack{\uparrow \\ 2k}}{\sigma} \cdot \underset{\substack{\uparrow \\ 2m}}{\tau} = 2(k+m)$

If inverse $\sigma = \tau_1 \ldots \tau_{2k}$

$\sigma^{-1} = \tau_{2k} \ldots \tau_1$

We can define the sign of a permutation $\mathrm{sgn}: S_n \to \{1,-1\}$ by the formula.

$$\mathrm{sgn}(\sigma) = \begin{cases} 1 & \text{if } \sigma \text{ is even,} \\ -1 & \text{if } \sigma \text{ is odd.} \end{cases}$$

If we think of $\{1,-1\}$ as a group under multiplication it is easy to see that sgn is an homomorphism.

$\mathrm{sgn}^{-1}[\{1\}]$ is a subgroup of S_n.

↑
identity

consisting of all even permutations

↓

We call it the alternating group of n letters! A_n.

cosets and the theorem of lagrange.

<u>Main idea</u>: the order of a subgroup of a finite group G is a divisor of the order of G.

recall order of a group $|G|$ = # of elements.

<u>Example</u>: $\mathbb{Z}_6 = \{0,1,2,3,4,5\}$ $|\mathbb{Z}_6| = 6$.

$\langle 3 \rangle = \{0,3\}$ $|\langle 3 \rangle| = 2$.

All finite cyclic groups.

Let $(G, \circ, e)$ be a group and H be a subgroup. Then H induces 2 equivalence relations:

- $\sim_L$: $a \sim_L b$ ssi $a^{-1}b \in H$

- $\sim_R$ $a \sim_R b$ iff $ab^{-1} \in H$.

It is an equivalence relation:

$[a]_{\sim_L} = \{b \mid a \sim_L b\} = \{b \mid a^{-1}b \in H\}$

$= \{b \mid \exists h \in H \ a^{-1}b = h\} = \{b \mid \exists h \in H \ b = ah\}$

$= aH$.

As $\sim_L$ induces an equivalence relation on G

so it induces a partition $[a]_{\sim_L}$

<u>Definition:</u> let H be a subgroup of G, the subset $aH = \{ah \mid h \in H\}$ of G is the <u>left coset of H containing a</u>,

- The subset $Ha = \{ha \mid h \in H\}$ of G is the <u>right coset</u> of H containing a.

e.g. $A = 3\mathbb{Z}$ $G = \mathbb{Z}$,

$3\mathbb{Z} = \{\ldots, -9, -6, -3, 0, 3, 6, 9, \ldots\}$

$1+3\mathbb{Z} = \{\ldots, -8, -5, -2, 1, 4, 7, 10, \ldots\}$

$2+3\mathbb{Z} = \{\ldots, -4, -1, 2, 5, 8, 11, \ldots\}$

<u>RmK:</u> For a subgroup H of an abelian group G, the partition into left cosets of H and the partition into right cosets are the same.

Another example: The group $\mathbb{Z}_6=\{0,1,2,3,4,5\}$ is abelian. The subgroup $H=\{0,3\}$

We find all the cosets of H in $\mathbb{Z}_6$.

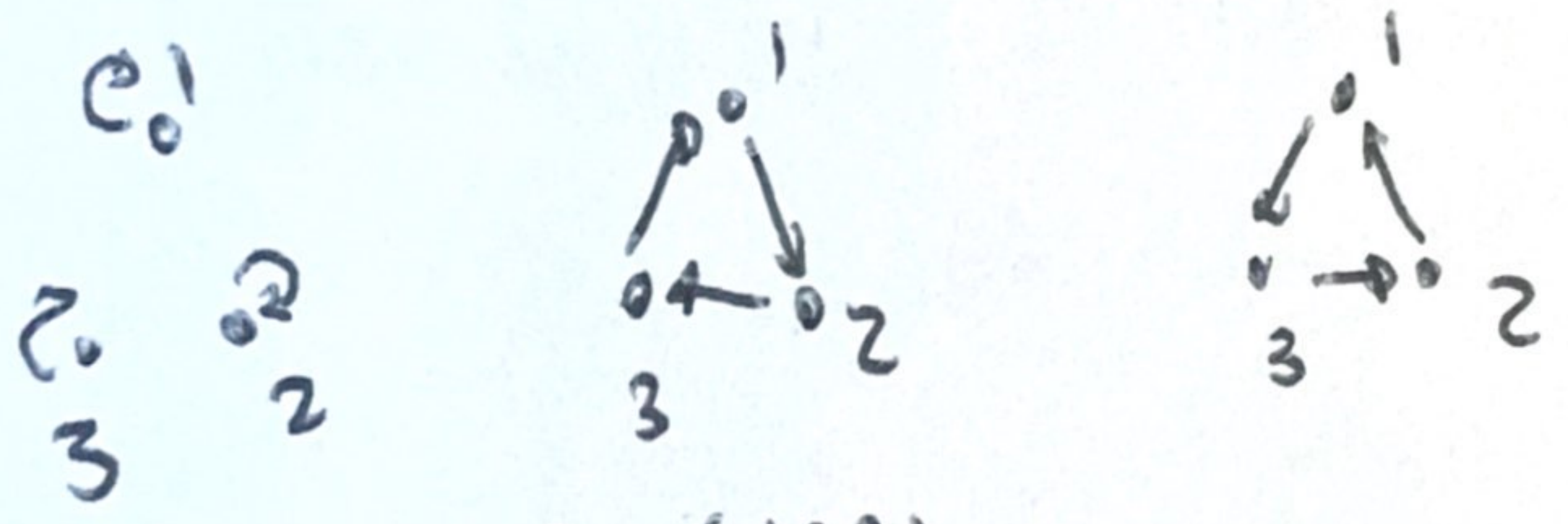
$H=\{0,3\}$

$1+H=\{1,4\}=H+1$

$2+H=\{2,5\}=H+2$.

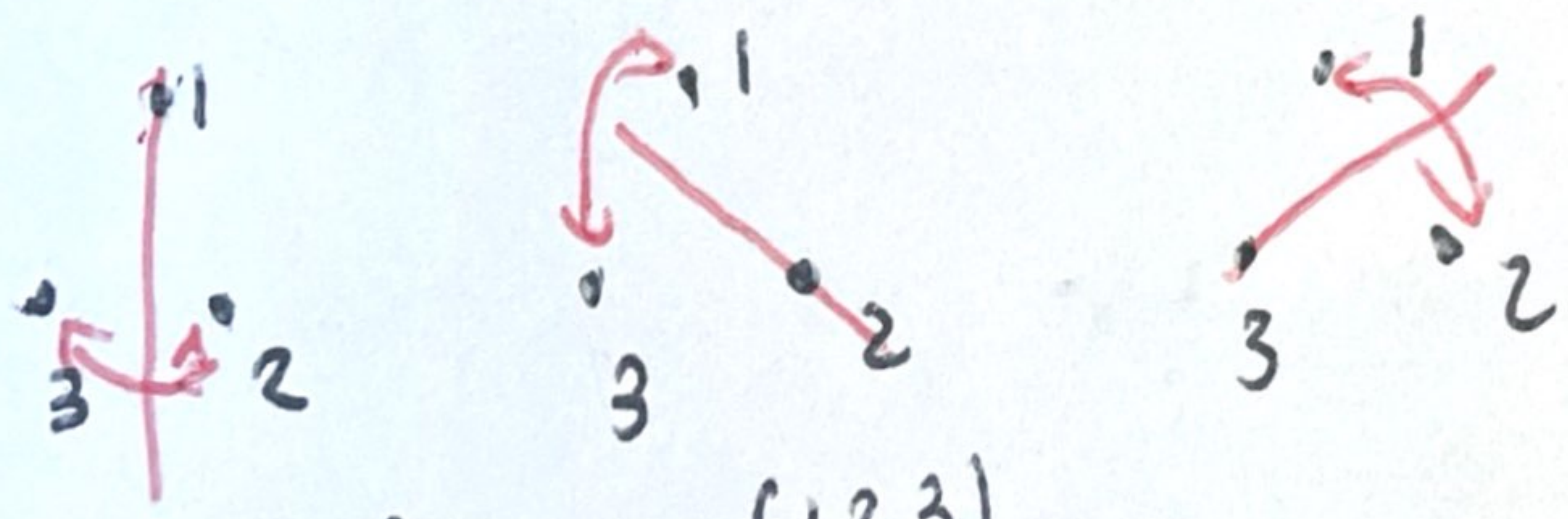
What is the group is not abelian? e.g. $G=S_3$

e:



$\begin{pmatrix}1&2&3\\3&1&2\end{pmatrix}=\rho_2$

$\rho_0=\begin{pmatrix}1&2&3\\1&2&3\end{pmatrix}$ $\rho_1=\begin{pmatrix}1&2&3\\2&3&1\end{pmatrix}$



$\mu_3=\begin{pmatrix}1&2&3\\2&1&3\end{pmatrix}$

$\mu_1=\begin{pmatrix}1&2&3\\1&3&2\end{pmatrix}$ $\mu_2=\begin{pmatrix}1&2&3\\3&2&1\end{pmatrix}$

$\rho_1\circ\mu_1=\begin{pmatrix}1&2&3\\2&1&3\end{pmatrix}$

$H=\langle\mu_1\rangle=\{\rho_0,\mu_1\}$

$\rho_1 H=\{\rho_1\circ\rho_0,\rho_1\circ\mu_1\}=\{\rho_1,\mu_3\}$

$\rho_2 H=\{\rho_2\rho_0,\rho_2\circ\mu_1\}=\{\rho_2,\mu_2\}$

La partición en cosets por la derecha es

$H = \{\rho_0, \mu_1\}$

$H\rho_1 = \{\rho_0\rho_1, \mu_1\rho_1\} = \{\rho_1, \mu_2\}$

$H\rho_2 = \{\rho_0\rho_2, \mu_1\rho_2\} = \{\rho_2, \mu_3\}$

The theorem of lagrange:

Let H be a subgroup of G.

$\boxed{\text{Key point}}$ every left coset (right coset) has the same number at elements of H.

~~if H is fi~~

↳ i.e. $\exists f: H \to aH$ that is a bijection.

$$\varphi_a: H \to aH$$
$$h \to a\cdot h$$

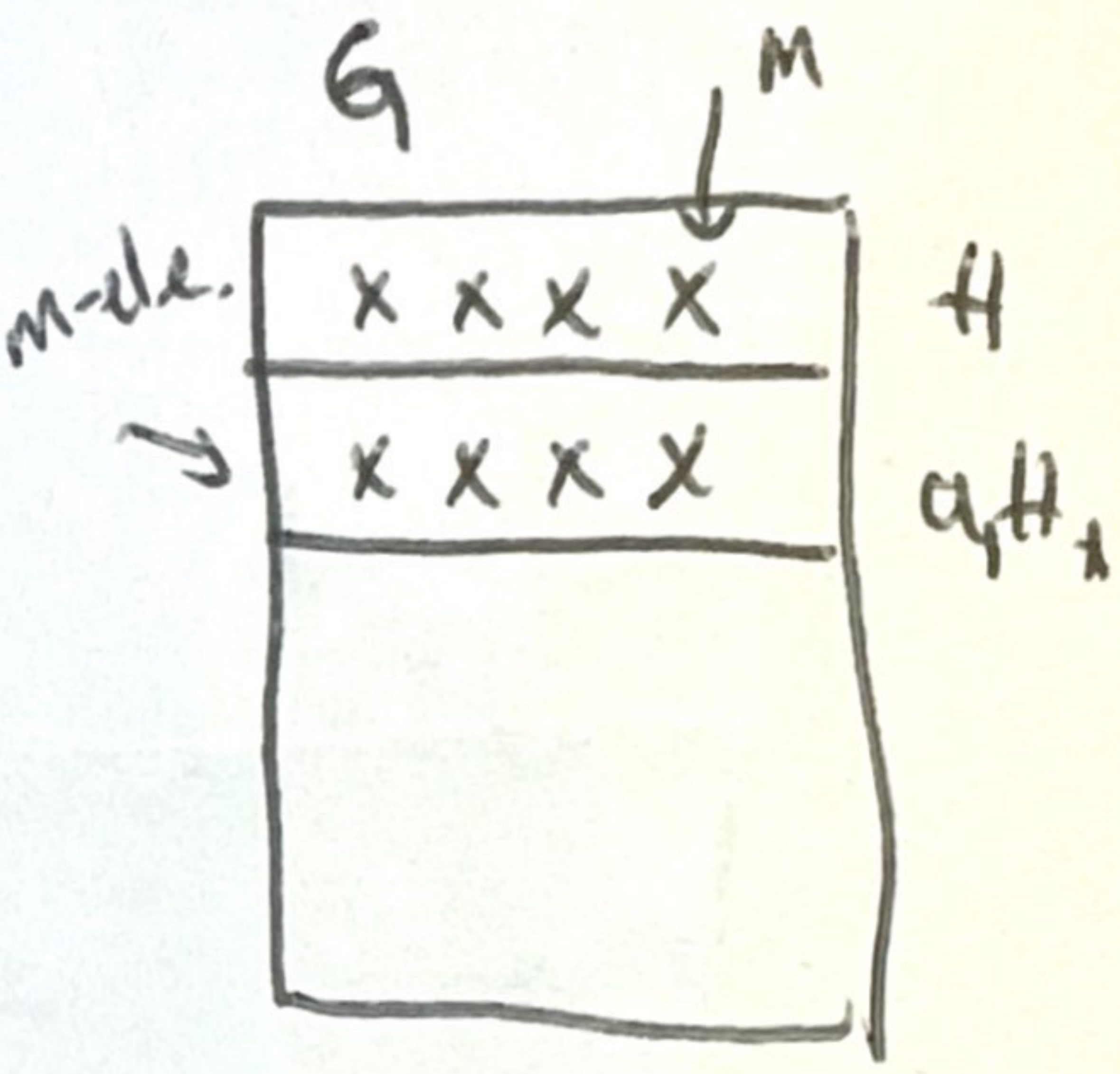
$\boxed{\text{is a bijection}}$:

clearly it is surjective. For injectivity, let $h_1, h_2 \in H$ s.t. $a\cdot h_1 = a\cdot h_2$ then by cancellation law $\boxed{h_1 = h_2}$

$\boxed{\text{Thm}}$: (Lagrange) Let H be a subgroup of a finite group G. Then the order of H is a divisor of the order of G.

Proof: Sea n el orden de G y supongamos que $|H| = m$. Every coset $|aH| = m$.

$$G = \dot{\bigcup_{i=1}^{r}} a_i H$$



$$|G| = \sum_{i=1}^{r} |a_i H| = r \cdot m$$

$$\Rightarrow n = rm.$$

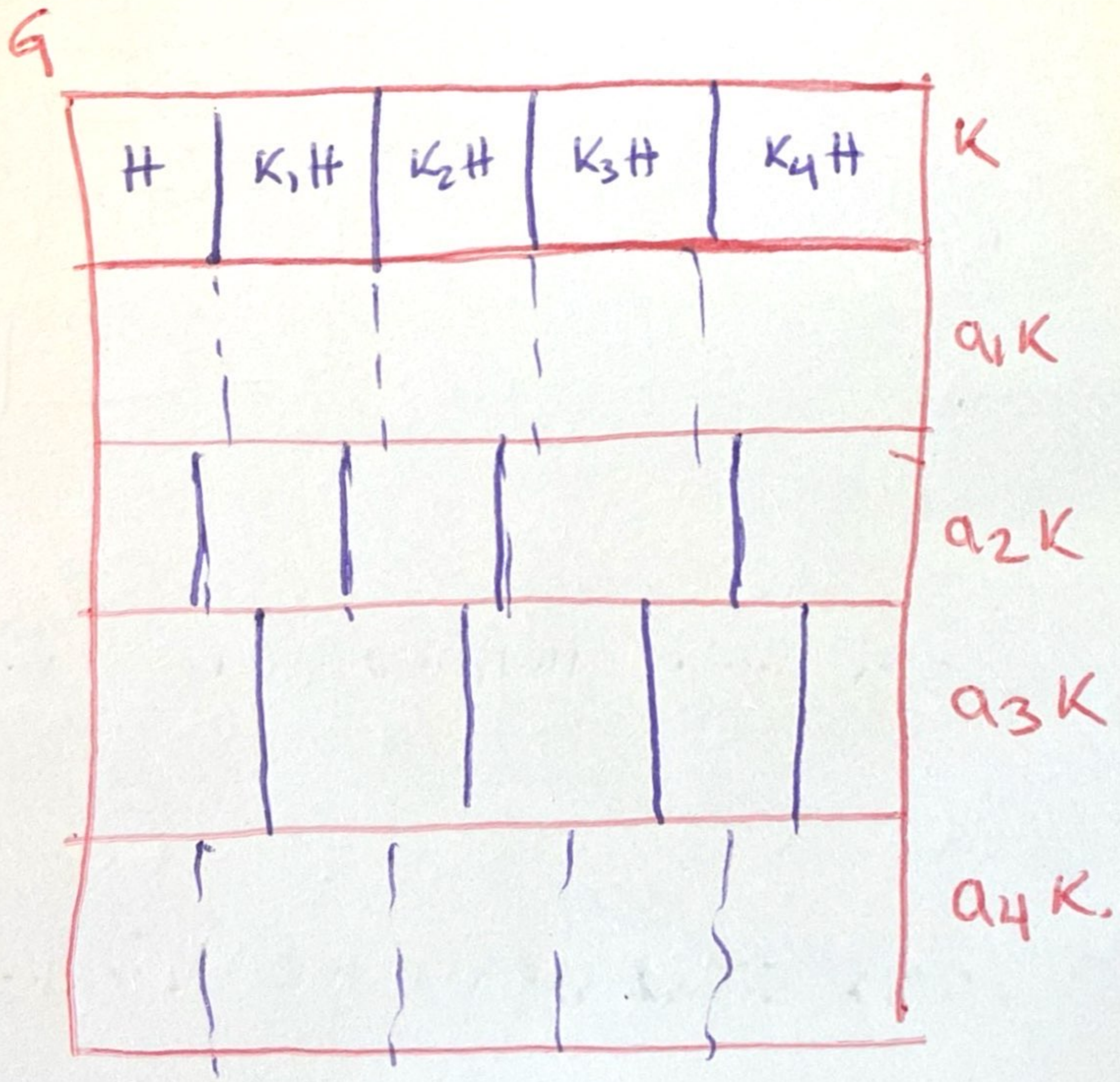
Corollary: every group of prime order is cyclic.

Proof: Let G be a group of prime order and let $a \in G$ $a \neq e_G$.

consider $H = \langle a \rangle$ by the thm $H = G$.

Definition: Let H be a subgroup of G. the number of left cosets of H is said the index of H in G (and it is $[G:H]$).

Thm: Suppose $H \leq K \leq G$ And assume $[G:K]$ $[K:H]$ are both finite. Then $[G:H]$ is finite and $[G:H] = [G:K] \cdot [K:H]$.



$|K| = r_1 \cdot |H|$

$|G| = r_2 \cdot |K|$

$|G| = r_2 \cdot r_1 \cdot |H|$

$\boxtimes$

Direct products and finitely generated abelian groups.

Pile of groups:

Finite $\begin{cases} \cdot\ \mathbb{Z}_n \\ \cdot\ S_n \\ \cdot\ A_n. \\ \cdot\ (U_n, \cdot) \end{cases}$

Infinite: $\begin{cases} \mathbb{Z} \\ \mathbb{R} \\ \mathbb{C} \end{cases}$ S_A for A infinite.

Goal: We want to build more examples of groups out of those that we already know.

Definition:

a) cartesian product of sets: let $S_1, \dots, S_n$ sets of order n-tuples $(a_1, \dots, a_n)$ where $a_i \in S_i$ for $i = 1, 2, \dots, n$. the cartesian product

$S_1 \times \dots \times S_n$ is denoted as $\prod_{i=1}^{n} S_i$

b) let $G_1, \dots, G_n$ groups then $\prod_{i=1}^{n} G_i$ is a group equipped with the following binary operation:

$$\cdot : \prod_{i=1}^{n} G_i \times \prod_{i=1}^{n} G_i \longrightarrow \prod_{i=1}^{n} G_i$$

$$\left((a_i)_{i=1}^{n}, (b_i)_{i=1}^{n}\right) \longrightarrow (a_i *_i b_i)_{i=1}^{n}$$

and we call it the $\boxed{\text{DIRECT PRODUCT OF GROUPS}}$

$\boxed{\text{Claim 1}}$ $\left(\prod_{i=1}^{n} G_i, \cdot\right)$ is a group if $(G_i, *_i, e_i)$ is a group.

- associative: $\left((a_1, \dots, a_n) \cdot (b_1, \dots, b_n)\right) \cdot (c_1, \dots, c_n)$

$= \left((a_1 *_1 b_1, \dots, a_n *_n b_n) \cdot (c_1, \dots, c_n)\right)$

$= \left((a_1 *_1 b_1) * c_1, \dots, (a_n *_n b_n) * c_n\right)$

$= \left(a_1 *_1 (b_1 * c_1), \dots, a_n *_n (b_n *_n c_n)\right)$

$= (a_1, \dots, a_n) \cdot (b_1 *_1 c_1, \dots, b_n *_n c_n)$

$= (a_1, \dots, a_n) \cdot \left((b_1, \dots, b_n) \cdot (c_1, \dots, c_n)\right)$

identity: $(e_1, \ldots, e_n) = e$.

inverse: $(a_1, \ldots, a_n)$ is $(a_1^{-1}, \ldots, a_n^{-1})$

When the operation is commutative sometimes we use the additive notation $\bigoplus_{i=1}^{n} G_i$

Example: $\mathbb{Z}_2 \times \mathbb{Z}_3$ $\quad \mathbb{Z}_2 = \{0,1\}$

$\uparrow$ $\quad \mathbb{Z}_3 = \{0,1,2\}$

cyclic!

$(1,1)$

$2(1,1) = (1,1) + (1,1) = (0,2)$

$3(1,1) = (1,1) + (1,1) + (1,1) = (1,0)$

$4(1,1) = (1,1) + (1,1) + (1,1) + (1,1) = (0,1)$

$5(1,1) = (1,1) + \cancel{(1,1)} + (1,1) + (1,1) + (1,1) = (1,2)$

$6(1,1) = (0,0)$

while $\mathbb{Z}_3 \times \mathbb{Z}_3$ is not cyclic!

in $\mathbb{Z}_3$ every element added to itself 3 times! is the identity. $\Rightarrow$ the same is true in $\mathbb{Z}_3 \times \mathbb{Z}_3$ $\rightarrow$ it has 9 elements

but no element can generate the group, since the group generated will have 3 elements.

Theorem: the group $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic and isomorphic to $\mathbb{Z}_{mn}$ iff HCF$(m,n)=1$.

Proof: Assume $(m,n)=1$.

- $\mathbb{Z}_m \times \mathbb{Z}_n$ has $m \cdot n$-elements.
- $\langle(1,1)\rangle$ is the group generated by adding $(1,1)$ to itself repeatetly.
- The first component becomes 0 after m summands, $2m$ summands, etc.
- the second component becomes 0 after n summands, $2n$ summands, etc.

$\Rightarrow$ to give $(0,0)$ in both coordinates it must be a multiple of both n and m i.e. it is ~~[illegible]~~ n and m -divisible. the order of $(1,1)$.

The first number that is divisible by n and m is nm if $(n,m)=1$.

$\Rightarrow$ $|\langle(1,1)\rangle| = nm$ so is the whole group since it is cyclic and it has mn-elements

$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$ by the isomorphism theorem of cyclic groups.

"$\Rightarrow$" we proceed by contrapositive.

Assume that ~~the~~ $GCD(n,m) = d$

Then $\frac{mn}{d}$ ~~divides~~ is divisible by both m and n.

<u>why?</u> $n = d \cdot q_1$ $\quad \frac{n \cdot m}{d} = d\, q_1 \cdot q_2$

$m = d\, q_2$

Then given $(r,s) \in \mathbb{Z}_m \times \mathbb{Z}_n$ we have:

$$\underbrace{(r,s) + (r,s) + \dots + (r,s)}_{\frac{mn}{d}\text{-summands}} = (0,0)$$

$\Rightarrow$ it is not cyclic since it is not generated by a unique element. $\Rightarrow \ncong \mathbb{Z}_{mn}$ since the later one is cyclic.

<u>Corollary:</u> $\prod_{i=1}^{n} \mathbb{Z}_{m_i}$ is cyclic ~~iff~~ and isomorphic to $\mathbb{Z}_{m_1 \dots m_n}$ iff $\gcd(m_i, m_j) = 1 \;\forall\, i \neq j$.

In particular, given n a natural number by the fundamental theorem of arithmetic

$$n = p_1^{r_1} \cdots p_k^{r_k}$$

Then $\mathbb{Z}_n \cong \prod_{i=1}^{k} \mathbb{Z}_{p_i^{r_i}}$

e.g. → thus $\mathbb{Z}_{72} \cong \mathbb{Z}_8 \times \mathbb{Z}_9$.

Remark: if we change the order of factors we obtain the same group.

Definition: Let $r_1, \dots, r_k$ be positive integers. Their least common multiple (lcm for short) is the smallest number k that is divisible by each of the r_i.

$\boxed{\text{Theorem}}$ Let $(a_1, \dots, a_k) \in \prod_{i=1}^{k} G_i$. If a_i is of order r_i in G_i then the order of $(a_1, \dots, a_k)$ is equal to the least common multiple of all the r_i.

Proof: For a power of $(a_1, \dots, a_k)$ to give the identity. The power must be a multiple of each of the r_i. The smallest number such that this happens is ~~lcd~~ $\operatorname{lcm}(r_1, \dots, r_k)$

Example: What is the order of $(8,4,10)$ in

$\mathbb{Z}_{12} \times \mathbb{Z}_{60} \times \mathbb{Z}_{24}$.

$12 = 2^2 \cdot 3$ $\cancel{\text{lcm}(12,60,24) = 2^3 \cdot 3 \cdot 5}$

$60 = 2^2 \cdot 3 \cdot 5$

$24 = 2^3 \cdot 3$

1) 8 has order $\cancel{4}\ 3$ in $\mathbb{Z}_{12}$ since $\dfrac{12}{\text{HCD}(8,12)} = \dfrac{12}{4} = 3$

2) 4 has order 15 in $\mathbb{Z}_{60}$ since $\dfrac{60}{4} = 15$

3) 10 has order 12 in $\mathbb{Z}_{24}$ since $\dfrac{24}{\text{HCD}(24,10)} = \dfrac{24}{2} = 12$

$\text{lcm}(12,15,3) = 2^2 \cdot 3 \cdot 5$ $\quad 12 = 2^2 \cdot 3$

$= \boxed{60}$ $\quad 15 = 3 \cdot 5$

$3 = 3$

$\boxed{(8,4,10) \text{ has order } 60 \text{ in } \mathbb{Z}_{12} \times \mathbb{Z}_{60} \times \mathbb{Z}_{24}}$

Remark: each G_i is a subgroup of $\prod_{j=1}^{k} G_j$

and it is seen as $(\underbrace{e_1, \dots, e_{i-1}}, g, \underbrace{e_{i+1}, \dots, e_k})$

element in G_i varying.

identity elements

THE STRUCTURE OF FINITELY GENERATED GROUPS

THM: [Fundamental theorem] Every finitely generated abelian group G is isomorphic to a direct product of cyclic groups i.e. G is isomorphic to a group of the form.

$$\mathbb{Z}_{p_1^{r_1}} \times \dots \times \mathbb{Z}_{p_k^{r_k}} \times (\mathbb{Z} \times \dots \times \mathbb{Z})$$

where the p_i are the prime numbers, the r_i are integers.

Betty number of G: # of factors of $\mathbb{Z}$

Rmk: • # of factors of prime powers $p_i^{r_i}$ is unique

and the Betty number is unique.

Some applications:

Definition: A group G is decomposable if it is isomorphic to a direct product of 2 proper nontrivial subgroups. Otherwise is indecomposible.

Thm: An abelian finite group is indecomposible iff it is cyclic ~~on~~ with order of a prime power.

Proof: $\Rightarrow$ $G \cong \mathbb{Z}_{p_1^{r_1}} \times \dots \times \mathbb{Z}_{p_k^{r_k}} \times (\mathbb{Z} \times \dots \times \mathbb{Z})$

since it is indecomposible $\rightarrow$ ~~no~~ only one $\mathbb{Z}_{p_i^{r_i}}$ occurs.

since it is finite $\rightarrow$ no $\mathbb{Z}$-factor occurs.

$\Leftarrow$ If $G \cong \mathbb{Z}_{p_i^{r_i}}$ and

~~$\mathbb{Z}_{p_i^{r_i}} \cong \mathbb{Z}_{p_i^{m}} \times \mathbb{Z}_{p_i^{m_2}} \rightarrow$ any element would have order~~

has to be a power of $\rightarrow$ p by lagrange.

$\mathbb{Z}_{p^{r_i}} \cong \mathbb{Z}_{p^m} \times \mathbb{Z}_{p^k}$ with $m+k=r_i$

any element would have order at most $p^{\max\{m,k\}} < p^{r_i}$.

thm: If m divides the order of a finite abelian group G then G has a subgroup of order m.

By the theorem

$$G \cong \mathbb{Z}_{p_1^{r_1}} \times \dots \times \mathbb{Z}_{p_k^{r_k}} \times \mathbb{Z} \times \dots \times \mathbb{Z}$$

since it is finite no $\mathbb{Z}$-component appears

$$G \cong \mathbb{Z}_{p_1^{r_1}} \times \dots \times \mathbb{Z}_{p_k^{r_k}}$$

where not all p_i's must be distinct.

The order of G is $p_1^{l_1} \cdot \ldots \cdot p_k^{l_k}$ where $0 \le l_i \le r_i$.

Then $p_i^{(r_i - l_i)}$ generates a cyclic subgroup of $\mathbb{Z}_{p_i^{r_i}}$ of order equal to the quotient of

$$\frac{p_i^{r_i}}{HCD(p_i^{r_i}, p_i^{r_i - s_i})} = p_i^{r_i - s_i}$$

$\Rightarrow$ it has order $\dfrac{p_i^{r_i}}{p_i^{r_i - s_i}} = p_i^{s_i}$

Recalling that $\langle a \rangle$ denotes the cyclic subgroup generated by a we see.

$$\langle (p_1)^{r_1 - s_1} \rangle \times \langle (p_2)^{r_2 - s_2} \rangle \times \dots \times \langle p_n^{r_n - s_n} \rangle.$$

is the required subgroup of order m